

Phase 1 Checklist — Infrastructure & “Pure Output” Sanity

This checklist is the **Definition of Done** for Phase 1 experiments.

Scope (Phase 1 experiments):

- E024 — Ulam spiral
- E124 — Klauber triangle
- E125 — Sacks spiral
- E126 — Hexagonal number spiral
- E019 — Prime density (PNT-style visualization; Phase-1 sanity level)

Goal: Ensure each experiment reliably produces correct artifacts, readable figures, and a short but complete report — **before** deep math work.

How to use

- Create a copy of this checklist per experiment (or keep one checklist and tick items per experiment).
 - Work in this order: **A** → **B** → **C** → **D** → **E** → **F**.
 - If something fails, fix earlier sections first (A/B) before polishing (C/D/E).
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A) Run & artifacts (output contract)

- ☐ A1. Experiment runs end-to-end without manual intervention.
- ☐ A2. Output directory is **only** `out/e###/` (no stray files elsewhere).
- ☐ A3. `out/e###/params.json` is written.
- ☐ A4. `out/e###/report.md` is written.
- ☐ A5. `out/e###/figures/` exists.
- ☐ A6. At least **one** figure PNG is saved into `out/e###/figures/`.
- ☐ A7. Figure filenames are stable and meaningful (no random hashes).
- ☐ A8. If writing fails (figure/report), the run **fails loudly** (non-zero exit / exception).
- ☐ A9. (If used by your pipeline) `out/e###/manifest.json` is written/updated.

Done for A: After one run, `out/e###/` is complete and consistent.

B) Parameters & reproducibility

- ☐ B1. `params.json` contains **all** parameters that affect output or runtime (e.g., `n_max`, `x_max`, `grid_type`, `layers`, `size`, `step`, `sample_size`).
- ☐ B2. If randomness exists, `seed` is a parameter and is saved to `params.json`.
- ☐ B3. Defaults are CI-safe (runtime stable; no accidental “huge run”).
- ☐ B4. Optional but recommended: two presets exist - `preset="ci"` (small/fast) - `preset="local"` (bigger/more detailed)

- ☐ B5. Re-running with the same params produces the same artifact structure (and for deterministic plots, the same images).

Done for B: Runs are comparable and reproducible; CI stays stable.

C) Figure quality (readability)

- ☐ C1. Each figure has a clear title (include key parameters like `n_max`, `x_max`, `size`, `layers`).
- ☐ C2. Axes are labeled **or** intentionally omitted and explained in the report (common for spirals).
- ☐ C3. Scale is explicit (linear/log clearly indicated).
- ☐ C4. Legends are present only if they add information (otherwise remove).
- ☐ C5. Overplotting is controlled (marker size/alpha/downsampling so patterns remain visible).
- ☐ C6. Export resolution is sufficient for both thumbnail and full-size viewing.
- ☐ C7. No clipped labels/titles (tight layout / bbox handling).

Done for C: A reader can understand what the plot is showing without extra explanation.

D) report.md minimum standard (Phase 1 level)

- ☐ D1. **Summary** (3–6 sentences): what is computed and what is shown.
- ☐ D2. **Parameters** section: bullet list of the main parameters and values.
- ☐ D3. **Key observation**: 2–5 concrete bullets that a reader can verify from the figures.
- ☐ D4. **Interpretation**: 1 short paragraph with cautious interpretation (no overclaim).
- ☐ D5. **Caveats**: at least 4 bullets, e.g. - finite N / finite range - sampling bias (linear vs log grids) - visualization artifacts / perception bias - runtime depends on implementation details
- ☐ D6. Use “observed up to ...” instead of “always/proves ...” unless it’s truly proved.
- ☐ D7. If multiple figures exist: one-line guide (“Fig 1 shows... Fig 2 shows...”).

Done for D: The report is short, complete, and safe in its claims.

E) Docs page & consistency

- ☐ E1. Experiment docs follow the same structure as your chosen reference page (e.g., e024): Goal → Method → Results → Interpretation → Caveats → References → Related experiments.
- ☐ E2. Tags are valid (restricted to `docs/tags.md`).
- ☐ E3. **Related experiments** lists 2–6 relevant experiment IDs/links.
- ☐ E4. References are minimal and useful (1–3 strong sources are enough).
- ☐ E5. No broken internal links (figures/reports/references).

Done for E: Docs build cleanly and pages look uniform.

F) Phase 1 batch sanity (project-level)

- ☐ F1. All Phase 1 experiments run back-to-back locally (E024, E124, E125, E126, E019).
- ☐ F2. Tests + lint/format + typing targets pass after the changes.
- ☐ F3. Docs build passes (no new warnings/errors introduced).

- ☐ F4. Gallery thumbnails look reasonable (at least one “hero” figure per experiment is readable).
- ☐ F5. No experiment writes outside its own `out/e###/` directory.

Done for F: Phase 1 is closed: reproducible, CI-safe, output-clean.

Notes (optional)

- Phase 1 is intentionally “low math knowledge”. Prefer clarity, reproducibility, and disciplined reporting.
- Deeper theory upgrades happen in Phase 2 (probabilistic guarantees, bounds, residue classes, etc.).